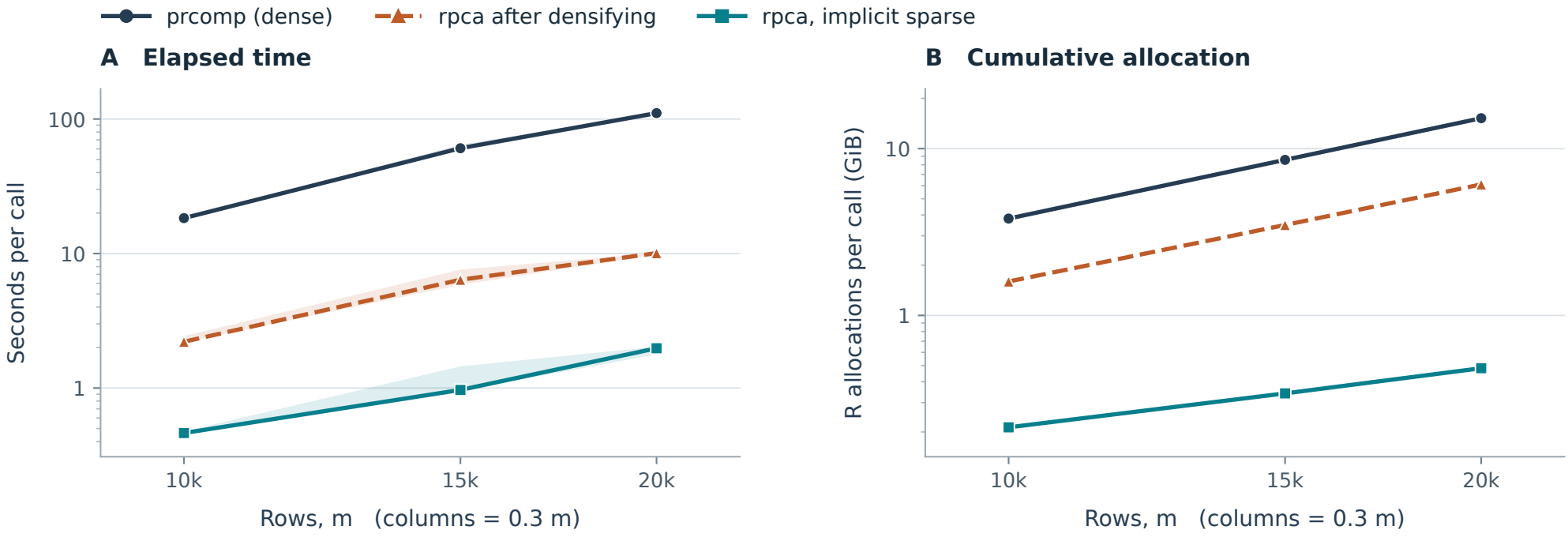
PCA on sparse input: preserving sparse storage

1.5–6 million nonzeros (5%); same preprocessing and 40 score columns



k = 40, p = 10, q = 2 · 5-run timing medians, IQR shaded · allocations: one profile · BLAS threads: 4

Largest observed error / optimal rank-40 error: 1.0000. R allocations exclude setup and do not measure peak RSS.